

BRIEF A — TECHNICAL MAINTENANCE ENGINEERING

Objective: detect pumps at risk of failure within the next 7 days using vibration, temperature, pressure, acoustic, motor current, and error-event sensor data, so maintenance can be scheduled proactively rather than reactively.

Model & Feature Engineering

Logistic Regression trained on 12 features: 6 raw sensor channels plus 6 engineered signal-health indicators — 6-reading rolling mean, RMS, peak-to-peak range, rolling standard deviation, and short-term trend slope, all computed per-asset on vibration. Features were standardised (zero mean, unit variance) before training, since Logistic Regression is sensitive to feature magnitude.

Accuracy	94.2%	Overall correct classification rate on held-out test data.
Sensitivity (recall)	84.3%	Share of true near-failures the model actually catches. This is the number to optimise — a m
False alarm rate	2.4%	Healthy pumps incorrectly flagged for inspection. Low enough that technician time isn't wasted
Shift t-test p-value	0.0026	Night-shift vibration is significantly higher than day-shift (mean 2.25g vs 2.15g) — set shift-awa
Depot ANOVA p-value	<0.0001	Mean pressure differs significantly by depot (114.1–118.8 psi range) — likely reflects real eleva
RUL model MAE	105.9 hours	Average error when predicting hours of remaining useful life via Linear Regression on the sar

Sensor Logic & Alert Thresholds

The rolling mean smooths transient sensor noise to reveal true degradation trend; RMS captures total vibration energy (rises with bearing wear or imbalance); peak-to-peak range flags erratic spikes; the new trend-slope feature detects whether vibration is actively rising right now, which is a stronger leading indicator than a single elevated reading. Recommended deployment: use the classifier as a daily fleet-wide triage pass, then run the RUL regression only on flagged pumps to prioritise servicing order.

BRIEF B — EXECUTIVE SUMMARY

Unplanned pump failures interrupt throughput, trigger emergency overtime labour, and carry real safety risk. This model identifies which pumps are likely to fail within the next seven days, so maintenance can be scheduled before a breakdown happens — turning an expensive emergency repair into a planned, budgeted service visit.

What changes operationally?	Maintenance shifts from a fixed calendar schedule to a risk-based one — technician pump
What does the model actually catch?	Roughly 8 in 10 pumps that are genuinely about to fail get flagged in advance, givi
How often does it cry wolf?	Under 3 in 100 healthy pumps get incorrectly flagged — technician time is not wast false
Why fund it?	Avoided downtime, fewer emergency callouts, and better-planned spare-parts orderin

What's the near-term ask?	Approve a 90-day pilot on the highest-criticality pumps. Track avoided downtime
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REFLECTION — SWITCHING CONTEXTS

1. Writing these two briefs from the same underlying model surfaced a real discipline gap between audiences. The technical brief could state a number and let it speak for itself — 'sensitivity 84.3%' is meaningful to an engineer without further explanation.
2. The executive brief required translating that same number into a consequence: not '84.3% sensitivity' but 'roughly 8 in 10 failures get caught in time.' The harder switch was resisting the urge to justify the model's limitations technically (why 84.3% and not higher) in the executive version — a CFO doesn't need the excuse, they need the number and what it costs or saves. Once I stopped trying to defend the model in both documents and instead asked 'what does this person need to decide?' for each audience separately, the two briefs wrote themselves fairly quickly.